

12(a)	total power of radiation emitted (by the star)	B1
12(b)	$F = \frac{L}{4\pi d^2}$ $= \frac{3.83 \times 10^{26}}{4 \times \pi \times 1.51 \times 10^{12}}$	C1
	$= 1340 \text{ W m}^{-2}$	A1

Question	Answer	Marks
12(c)	$m = \frac{E}{c^2}$ $= \frac{3.83 \times 10^{26}}{3.00 \times 10^{82}}$ $= 4.26 \times 10^9 \text{ kg}$	A1
12(d)	$L = 4\pi\sigma r^2 T^4$ $3.83 \times 10^{26} = 4 \times \pi \times 5.67 \times 10^{-8} \times 6.96 \times 10^{82} \times T^4 \text{ leading to } T = 5770 \text{ K}$	B1
12(e)	$\lambda_{(\text{max})} \propto \frac{1}{T}$ $\frac{5.00 \times 10^{-7}}{\lambda} = \frac{9940}{5770}$	C1
	$\lambda = 2.90 \times 10^{-7} \text{ m}$	A1